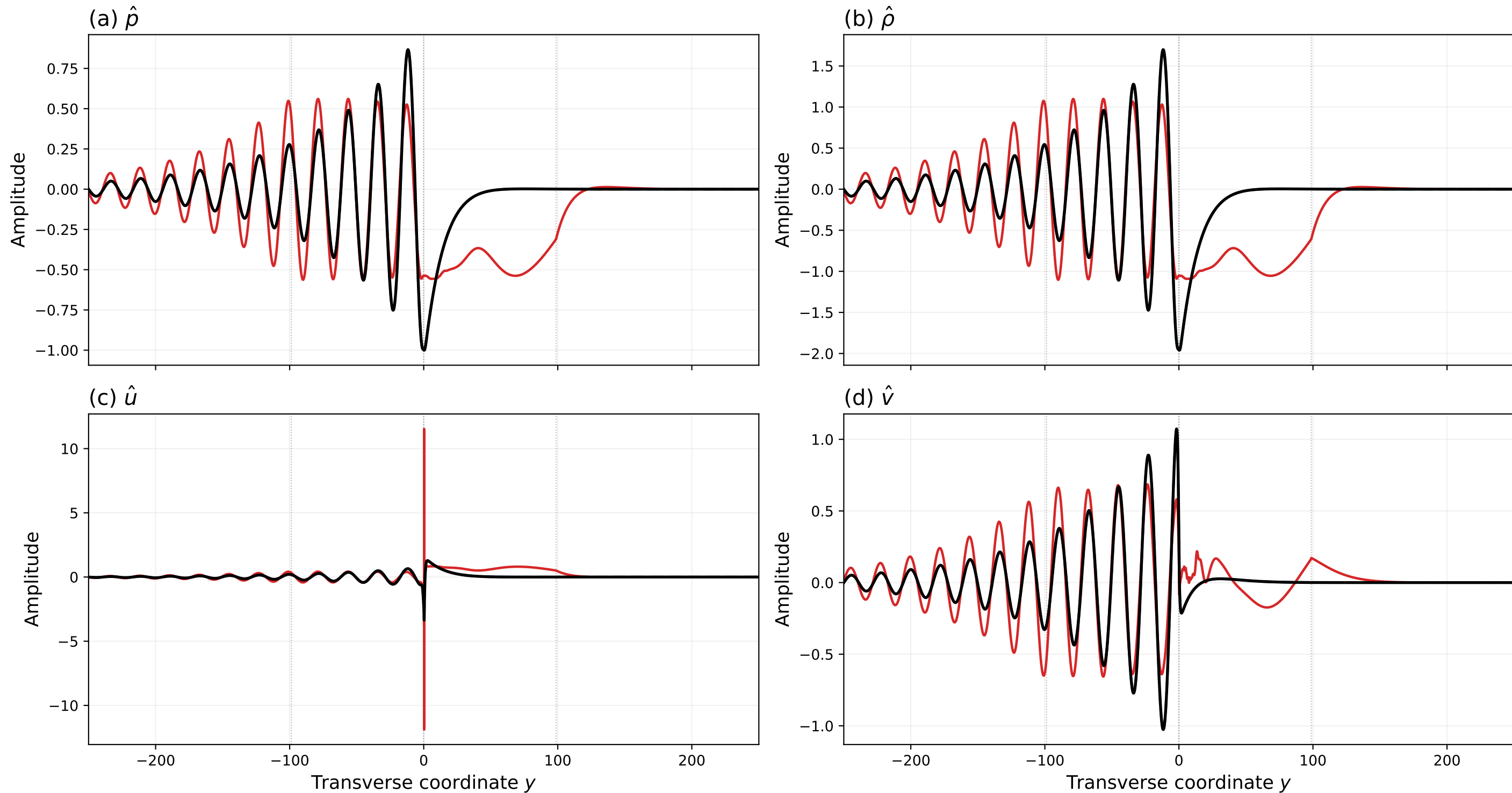


Imaginary parts of the representative supersonic mode: $M = 1.4$, $\alpha = 0.18$
 $c^* = 0.332906 + 0.043102i$, $c_{\text{PINN}} = 0.333275 + 0.042879i$, $c_{\text{shoot}} = 0.332904 + 0.043103i$

— Classical reference — PINN-seeded shooting — Direct PINN



Light dotted vertical lines mark the direct-PINN mapped-domain boundary; outside it, only the analytical far-field continuation is shown.